

Computational Economics

Course Introduction

Aims, syllabus, learning activities, and assessment

Yaolang Zhong yaolang.zhong@e.u-tokyo.ac.jp
Graduate School of Economics, The University of Tokyo

What Will You Learn?

Use computation to turn an economic model into results that you can explain and check.

- ▶ Translate a model into states, decisions, equations, and an algorithm.
- ▶ Choose numerical methods and understand their assumptions and approximations.
- ▶ Implement a calculation, check its output, and make it reproducible.
- ▶ Read computational research and connect its methods to your own questions.

How the Module Works

Activity	What you will do
Lectures	Build intuition, define the mathematical objects, and study computational methods
Labs	Work with practical tools, run examples, and verify results
Homework	Read, practice, and explain what you have learned
Final project	Apply computational methods through a research-paper replication

Each class has two parts of about 50 minutes. Weekly pages organize the schedule; topic-based decks may continue across sessions.

Syllabus at a Glance

Sessions	Topics
Opening	Course introduction; Markov processes and Markov decision processes
Core methods	Dynamic programming; Euler equations and endogenous grids; expectations; simulation
Approximation and learning	Function approximation; neural networks; supervised and unsupervised learning; reinforcement learning
Applications and synthesis	Heterogeneous-agent models; connections between methods
Research literature	Computational economics using machine-learning methods

See the [online syllabus](#) for the weekly schedule. Literature selections and the closing-session programme will be announced there.

Your Course Website

Open Computational Economics

- ▶ **Syllabus:** class dates and topic sequence.
- ▶ **Weekly materials:** lecture and lab slides, notebooks, and homework decks.
- ▶ **References:** reading resources and source links.
- ▶ **Grading:** assessment and replication-project requirements.

Bookmark the homepage. Use **Open in Colab** to run the introductory notebook online, or download it for VS Code.



Scan to open the course website.

Assessment and Grading

The module includes a replication project and weekly reading or exercise activities.

Assessment detail	Announcement
Grade weights	To be announced
Homework assessment	To be announced
Project team size and scope	To be announced
Deadlines and submission channel	To be announced

The confirmed assessment arrangements will appear in the syllabus and project guide.

The Replication Project

Choose a research paper and reproduce its computational or quantitative methods.

- ▶ Explain the research question, key equations, and algorithm.
- ▶ Organize runnable code with clear setup and execution instructions.
- ▶ Check your results and explain differences from the paper.
- ▶ Present your replication work in a **15-minute presentation**.

Presentations take place in the last one or two lecture weeks, depending on the number of submissions. See the [grading and project guide](#).

Our Opening Session

Time	Activity
First part: about 15 minutes	Course introduction: aims, website, syllabus, and assessment
First part: about 35 minutes	Introduction to Markov decision processes (MDP), beginning with Markov processes
Second part: about 50 minutes	Introduction to Git and Python; development tools and a first notebook

After class, use the separate [Setting Up Git and Python](#) homework deck for the setup exercise.

Practical Information and Questions

- ▶ **Teaching period:** October 7, 2026–January 27, 2027. Consult the online schedule for individual class dates.
- ▶ **Meeting:** Wednesdays, Period 4; International Academic Research Building, Seminar Room 514.
- ▶ **Contact:** Yaolang Zhong, yaolang.zhong@e.u-tokyo.ac.jp.
- ▶ **Office hours and consultation arrangements:** to be announced.

Questions about the course or preparation?

Next: examples and definitions of Markov processes.